

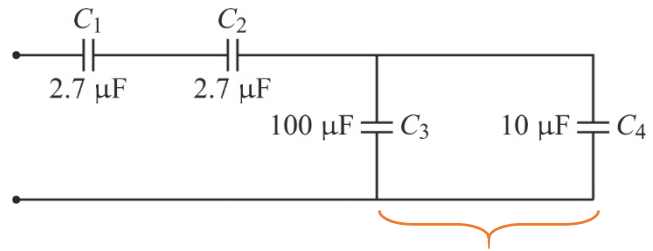
Exercise (3) – from Chapter 2. The complete list of problems is in the book.

1. Find the equivalent capacitance across the terminals of the circuit.

Solution:

$$\begin{aligned} C_1 &:= 2.7 \mu\text{F} & C_2 &:= C_1 \\ C_3 &:= 100 \mu\text{F} & C_4 &:= 10 \mu\text{F} \\ C_p &:= C_3 + C_4 = 110 \mu\text{F} \end{aligned}$$

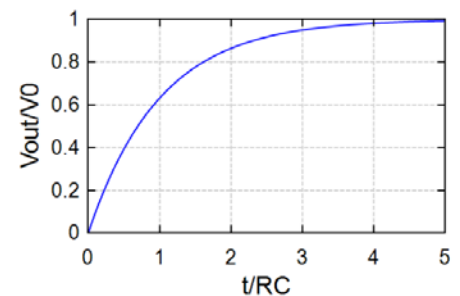
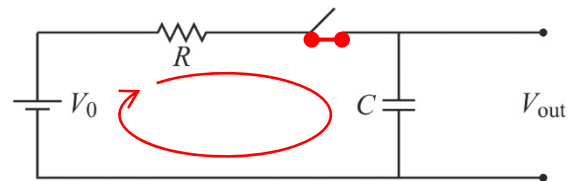
$$C_{tot} := \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_p} \right)^{-1} = 1.334 \mu\text{F}$$



2. Sketch V_{out} versus time for the circuit after the switch is closed. Assume the capacitor is initially uncharged and that $V_0 = 12 \text{ V}$, $R = 100 \text{ k}\Omega$, and $C = 10 \mu\text{F}$. Include appropriate numeric scales on the axes.

Solution:

$$V_{out} = V_0 \left(1 - e^{-\frac{t}{RC}} \right)$$



Appropriate scales are relative values of $\frac{V_{out}}{V_0}$ versus $\frac{t}{RC}$.

3. Referring to the previous circuit (*Problem-2.2*) and taking $V_0 = 100 \text{ V}$, $R = 1 \text{ M}\Omega$, what value of C is needed so that $V_{out} = 70 \text{ V}$ at 10.0 s after the switch closes?

Solution:

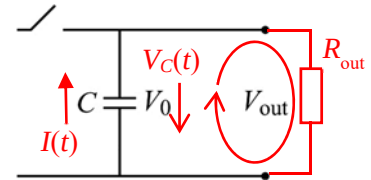
$$V_0 := 100 \text{ V} \quad R := 1 \text{ M}\Omega \quad V_{out} := 70 \text{ V} \quad t := 10 \text{ s}$$

Rearranging the charging equation:

$$1 - \frac{V_{out}}{V_0} = \exp\left(-\frac{t}{RC}\right) \rightarrow \ln\left(1 - \frac{V_{out}}{V_0}\right) = -\frac{t}{RC} \rightarrow \ln\frac{V_0}{V_0 - V_{out}} = +\frac{t}{RC} \rightarrow$$

$$C := \frac{t}{R \cdot \ln\left(\frac{V_0}{V_0 - V_{out}}\right)} = 8.306 \mu\text{F}$$

4. Suppose we use the circuit of *Problem-2.3* and wait a long time after the switch is closed so that the capacitor is fully charged. Now we open the switch and attach a 10 kΩ resistor across the output terminals. How long will it take for the voltage across the capacitor to drop to 1.0 V?



Solution:

$$V_{out} = V_C = V_0 e^{-\frac{t}{R_{out}C}}$$

Rearrange with the parameters given to calculate the time for voltage drop:

$$V_0 := 100 \text{ V} \quad R_{out} := 10 \text{ k}\Omega \quad V_{out} := 1 \text{ V}$$

$$\frac{V_{out}}{V_0} = \exp\left(-\frac{t}{R_{out}C}\right) \rightarrow \ln \frac{V_{out}}{V_0} = -\frac{t}{R_{out}C} \rightarrow (R_{out}C) \ln \frac{V_0}{V_{out}} = t \rightarrow$$

$$t := R_{out} \cdot C \cdot \ln\left(\frac{V_0}{V_{out}}\right) = 0.382 \text{ s}$$

5. A sine wave with **peak-to-peak** amplitude 20 V_{pp} is connected to a 10 kΩ resistor. Calculate the **peak**, the **rms**, and the **average** currents through the resistor. What power rating should the resistor have?

Solution:

First, summarize the definitions, then plug the values:

$$i(t) = I_p \sin(\omega t), \quad I_{pp} = 2I_p, \quad I_{ave} = \frac{\int_{\tau}^{\tau+T} i(t) dt}{T} = 0, \quad I_{rms} = \sqrt{\frac{\int_{\tau}^{\tau+T} i^2(t) dt}{T}} = \frac{I_p}{\sqrt{2}}, \quad P = I_{rms}^2 R$$

$$I_p := \frac{1}{2} \cdot \frac{V_{pp}}{R} = 1 \text{ mA} \quad I_{rms} := \frac{I_p}{\sqrt{2}} = 0.707 \text{ mA} \quad P := I_{rms}^2 \cdot R = 5 \text{ mW}$$

6. Determine the resonant angular frequency ω_0 for the given circuit.



Solution:

Apply the law of loops with complex impedances:

$$\tilde{I} = \frac{\tilde{V}_{in}}{\tilde{Z}_C + R + \tilde{Z}_L} = \frac{\tilde{V}_{in}}{\frac{1}{j\omega C} + R + j\omega L} = \frac{\tilde{V}_{in}}{R + j\left(\omega L - \frac{1}{\omega C}\right)} \rightarrow \frac{I_p}{V_{inp}} = \frac{1}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$

Set the derivative to zero to find its maximum:

$$0 = \frac{d}{d\omega} \left(\frac{I_p}{V_{inp}} \right) = \frac{1}{\left[R^2 + \left(\omega L - \frac{1}{\omega C} \right)^2 \right]^{3/2}} \cdot 2 \left(\omega L - \frac{1}{\omega C} \right) \left(L + \frac{1}{\omega^2 C} \right) \rightarrow \omega L = \frac{1}{\omega C} \rightarrow \omega_0 = \frac{1}{\sqrt{LC}}$$

$$C := 2 \text{ }\mu\text{F} \quad L := 1 \text{ H} \quad R := 1 \text{ k}\Omega$$

$$\omega_0 := \frac{1}{\sqrt{L \cdot C}} = 707.107 \frac{\text{rad}}{\text{s}} \quad f_0 := \frac{\omega_0}{2 \cdot \pi} = 112.54 \text{ Hz}$$

8. Design an RC low-pass filter that has $\frac{|\tilde{V}_{out}|}{|\tilde{V}_{in}|} = 0.5$ at 5 kHz.

Solution:

The low-pass filter is an RC divider with $V_{out} = V_C$ as shown. Apply the law of loops with complex impedances (as in 7):

$$0 = \tilde{V}_R + \tilde{V}_{out} - \tilde{V}_{in} \rightarrow \tilde{V}_{in} = R\tilde{I} + \tilde{Z}_C\tilde{I} \rightarrow \tilde{I} = \frac{\tilde{V}_{in}}{R + \tilde{Z}_C}$$

$$\tilde{V}_{out} = \tilde{V}_{in} \frac{\tilde{Z}_C}{R + \tilde{Z}_C} \rightarrow \frac{|\tilde{V}_{out}|}{|\tilde{V}_{in}|} = \left| \frac{\tilde{Z}_C}{R + \tilde{Z}_C} \right| = \frac{1/(\omega C)}{\sqrt{R^2 + 1/(\omega C)^2}} = \frac{1}{\sqrt{(\omega CR)^2 + 1}}$$

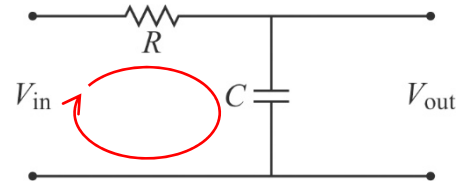
Solve for RC by squaring, inverting and rearranging:

$$\left(\frac{V_{out}}{V_{in}} \right)^2 = \frac{1}{(\omega CR)^2 + 1} \rightarrow \left(\frac{V_{in}}{V_{out}} \right)^2 = (\omega CR)^2 + 1 \rightarrow \omega CR = \sqrt{\left(\frac{V_{in}}{V_{out}} \right)^2 - 1}$$

Since only the product RC can be specified, we may choose one and calculate the other:

$$f := 5 \text{ kHz} \quad V_{out} := 0.5 \text{ V} \quad V_{in} := 1 \text{ V}$$

$$RC := \frac{\sqrt{\left(\frac{V_{in}}{V_{out}} \right)^2 - 1}}{2 \cdot \pi \cdot f} = 5.513 \cdot 10^{-5} \text{ s} \quad R := 1 \text{ k}\Omega \quad C := \frac{RC}{R} = 55.133 \text{ nF}$$



9. Derive the following expression for the circuit:

$$\frac{|\tilde{V}_{out}|}{|\tilde{V}_s|} = \frac{R_2}{\sqrt{(R_1 + R_2)^2 + (\omega R_1 R_2 C)^2}}$$

Solution:

The circuit is an RC divider with $V_{out} = V_C$, but R_2 is parallel to $\tilde{Z}_C$. Apply the law of loops as shown with complex impedances and $\tilde{Z}_2 = \tilde{Z}_{R_2 || C}$:

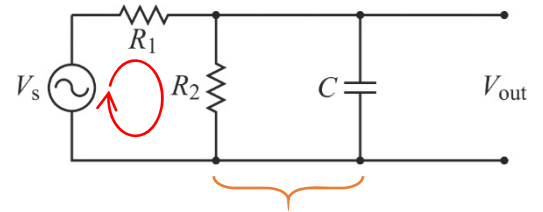
$$0 = \tilde{V}_{R_1} + \tilde{V}_{out} - \tilde{V}_s \rightarrow \tilde{V}_s = R_1\tilde{I} + \tilde{Z}_2\tilde{I} \rightarrow \tilde{I} = \frac{\tilde{V}_s}{R + \tilde{Z}_2}$$

$$\tilde{Z}_2 = \frac{R_2 \tilde{Z}_C}{R_2 + \tilde{Z}_C} = \frac{R_2 \frac{1}{j\omega C}}{R_2 + \frac{1}{j\omega C}} = \frac{R_2}{j\omega CR_2 + 1}$$

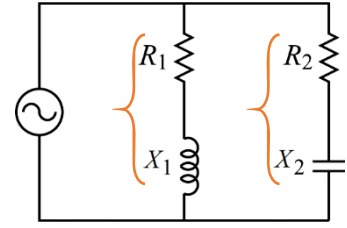
$$\tilde{V}_{out} = \tilde{V}_s \frac{\tilde{Z}_2}{R_1 + \tilde{Z}_2} = \frac{\frac{R_2}{j\omega CR_2 + 1}}{R_1 + \frac{R_2}{j\omega CR_2 + 1}} = \frac{R_2}{R_1(j\omega CR_2 + 1) + R_2} = \frac{R_2}{j(\omega CR_2 R_1) + (R_1 + R_2)}$$

We can now calculate the ratio of the magnitudes as a last step:

$$\frac{|\tilde{V}_{out}|}{|\tilde{V}_s|} = \frac{R_2}{\sqrt{(R_1 + R_2)^2 + (\omega R_1 R_2 C)^2}}$$



10. Consider the circuit, where $R_1 = 20 \Omega$, $X_1 = 37.7 \Omega$, $R_2 = 10 \Omega$, and $X_2 = -53.1 \Omega$. Compute the **rms** magnitude of the current flowing out of the signal generator and the phase angle between that current and the signal generator voltage. Assume the signal generator outputs a $230 V_{rms}$ sine wave with a frequency of 60 Hz.



Solution:

First we find the total impedance of the circuit. R_1 and jX_1 are in series, as are R_2 and jX_2 , so these may be simply added together. The two resulting impedances $\tilde{Z}_1 = R_1 + jX_1$ and $\tilde{Z}_2 = R_2 + jX_2$ are in parallel, thus:

$$\tilde{Z}_{tot} = \tilde{Z}_1 || \tilde{Z}_2 = \frac{\tilde{Z}_1 \tilde{Z}_2}{\tilde{Z}_1 + \tilde{Z}_2} = \frac{(R_1 + jX_1)(R_2 + jX_2)}{(R_1 + jX_1) + (R_2 + jX_2)} = \frac{(R_1 R_2 - X_1 X_2) + j(R_1 X_2 + R_2 X_1)}{(R_1 + R_2) + j(X_1 + X_2)}$$

To simplify, change both the numerator and the denominator to complex exponentials:

$$\tilde{Z}_{tot} = \frac{A_n e^{j\varphi_n}}{A_d e^{j\varphi_d}} = \frac{A_n}{A_d} e^{j(\varphi_n - \varphi_d)} = \frac{\sqrt{(R_1 R_2 - X_1 X_2)^2 + (R_1 X_2 + R_2 X_1)^2}}{\sqrt{(R_1 + R_2)^2 + (X_1 + X_2)^2}} e^{j(\varphi_n - \varphi_d)}$$

$$\varphi_n - \varphi_d = \text{atan}\left(\frac{R_1 X_2 + R_2 X_1}{R_1 R_2 - X_1 X_2}\right) - \text{atan}\left(\frac{X_1 + X_2}{R_1 + R_2}\right)$$

$$\tilde{I} = \frac{\tilde{V}}{\tilde{Z}_{tot}} = \frac{|\tilde{V}| e^{j0}}{|\tilde{Z}_{tot}| e^{j(\varphi_n - \varphi_d)}} = \frac{|\tilde{V}|}{|\tilde{Z}_{tot}|} e^{-j(\varphi_n - \varphi_d)} = \frac{V}{Z_{tot}} e^{j(\varphi_d - \varphi_n)}$$

Now we can put in the numbers:

$$V_{rms} := 230 \text{ V} \quad R_1 := 20 \Omega \quad X_1 := 37.7 \Omega \quad R_2 := 10 \Omega \quad X_2 := -53.1 \Omega$$

$$Z_{tot} := \frac{\sqrt{(R_1 \cdot R_2 - X_1 \cdot X_2)^2 + (R_1 \cdot X_2 + R_2 \cdot X_1)^2}}{\sqrt{(R_1 + R_2)^2 + (X_1 + X_2)^2}} = 68.382 \Omega$$

$$\varphi_n := \text{atan}\left(\frac{R_1 \cdot X_2 + R_2 \cdot X_1}{R_1 \cdot R_2 - X_1 \cdot X_2}\right) \quad \varphi_d := \text{atan}\left(\frac{X_1 + X_2}{R_1 + R_2}\right)$$

$$I_{rms} := \frac{V_{rms}}{Z_{tot}} = 3.363 \text{ A} \quad \varphi_d - \varphi_n = -9.892^\circ$$

The result can be verified by using complex calculus directly:

$$Z_{1\sim} := R_1 + i \cdot X_1 \quad Z_{2\sim} := R_2 + i \cdot X_2$$

$$I_{rms\sim} := V_{rms} \cdot \left(\frac{1}{Z_{1\sim}} + \frac{1}{Z_{2\sim}} \right) \quad |I_{rms\sim}| = 3.363 \text{ A} \quad \arg(I_{rms\sim}) = -9.892^\circ$$